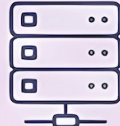


Experimental Inputs



4 Instruments

IPIP-NEO-120, SD3, ZKPQ-50-CC, EPQR-A
• 221 items, 17 domains, 63 reverse-keyed



20 LLMs (8 Vendors)

Anthropic, OpenAI, Google, DeepSeek,
Alibaba, Zhipu, Moonshot, MiniMax



17 Conditions

1 Default (Honest) +
16 MBTI Personas

Sampling Strategy

$k = 5$ samples per cell | Temperature $T = 0.7$

Six Standard Psychometric Checks



1. Internal Consistency (Cronbach's α)

FAIL
Spearman $\rho = -1.0$
with reverse density



2. Factor Structure (EFA)

FAIL
5 $\rightarrow$ 3 factor collapse
Tucker $\phi = 0.993$



3. Reverse-Item Consistency (PIR)

FAIL
PIR = 0.468
(High inconsistency)



4. Variance Decomposition

FAIL
Model Variance (SS) < 1%



5. Convergent Validity

FAIL
7/8 sign-match
but weak r values



6. Measurement Invariance

FAIL
0/320 pairs show
strong invariance

Persona Stress Test

Tested Variables: PSD (Persona Separation Degree)

- Nearest-Centroid Fidelity • Target Adherence • Item Plasticity
- Factorial MBTI Main Effects • Within-Sample Stability

99.7% nearest-centroid recovery
vs. 0.839 cross-scale coherence

COMPLIANCE WITHOUT VALIDITY ⚠